The Metabolic Module on the Amino Acids

The offline flux store carried to the Mülleder screen,

and where the endpoint's noise floor sits

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Abstract

The metabolic module built for gene essentiality (`notes-tex/028-gene-essentiality`) carried to Mülleder's 19 intracellular amino acids on the prototrophic deletion collection on minimal medium. The module is an offline store, one flux vector per deletion from a parsimonious solve on the prototroph with 19 synthesis maxima beside it, read through a zero-initialized affine on a 19-output head, and every arm is paired by seed against the same backbone without it under the 027 protocol on Merzbacher's pinned test genes. The endpoint sits at its noise floor: the screen's own strain-level retest rank is 0.20, the second genome-wide screen of the same phenotype (Cooper 2010) agrees with it at 0.03 over 4,051 shared strains and reproduces its own value for a typical gene at 0.05, and every model, transformer or ladder, at width 12 or 90, with or without a store, reads between 0.00 and 0.06. Four forms of the store affine sit at or under their control, the cached solve head recovers the affine's loss and lands at the control, and width, 100 more epochs and the rule counts change nothing. What the campaign fixes is the target's position rather than the module's. Every number is at a validation-selected epoch on the pinned test set with the across-seed spread, every table is generated from the result files, and the betaxanthin half of the campaign is the sibling document `notes-tex/031-betaxanthin-module`.

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1 The endpoint, the protocol and the stores ✗

Mülleder's screen measures 19 intracellular amino acids, in mM, on the prototrophic deletion collection (markers restored) grown on synthetic minimal medium, one culture per strain, 4,678 strains. Every cell is scored under the 027 protocol: Merzbacher's 639 test genes pinned to test, the epoch selected on validation Pearson (centered five-epoch moving average), and the mean per-amino-acid pinned-test Spearman over the 19 reported; the comparators are the 027 rounds' own numbers on the same split, read from their result files (Table 4). The backbone is the essentiality one (pathway multi-hot with ProtT5, width 12, one block, four attention heads, CRPS on the head, 200 epochs), a fifth of the width 027 trained, with a 19-output head on the mean channel. The graph regularizer is requested on every cell and never evaluated at width 12: it is pinned to the second block, and these cells have one, so `graph_reg_loss` is 0 at all 200 epochs. It is inert in the control and in the arm alike, so the pairing is unaffected, and every arm is paired by seed over six seeds against the same backbone with no affine. The screen's own strain-level rank retest is 0.20, the agreement of two measurements of one strain, which bounds a second screen; a model scored against one measurement is bounded by the square root of the reliability, 0.62 on the Pearson scale on average over the nineteen amino acids and about 0.45 on the Spearman scale (Section 4, Table 11).

The study began inside the essentiality campaign (rounds 118 onward there) and has been its own study, 032, since 2026-09-23, with its own results tree, W&B project and round series (A1 onward; the map from the round numbers used here to the A-series is in the study's note). The betaxanthin half of the same campaign is the sibling document, `notes-tex/031-betaxanthin-module`, and the essentiality endpoint the backbone and the store were built on is `notes-tex/028-gene-essentiality`.

The store represents the strain, not the wild type. For the amino acids (`aa_reference_flux_table.py`, job 2751) the model is the unmodified prototroph on the dataset's medium; per GEM gene one parsimonious growth solve and 19 linear programs, each amino acid's synthesis maximum at 90 percent of own growth, kept as summary columns beside the growth vector rather than as 19 further vectors. On the graph side the processor had removed the two allele genes and the BTS1 copy from every cell graph, which this strain never carried; round 124's `-deletion-only-graph` leaves the graph as the strain is.

2 The rounds ✗

Round 118 (jobs 2743 and 2744, w135) reads the essentiality store's growth vector on the amino-acid medium into the 19-output affine on the mean channel. Round 121 (w138) is the amino-acid production store, its 19 ratios and the growth ratio alone, and the whole vector with all 42 summary statistics. Round 122 (w139) is the affine-free control on the backbone 020 and 027 trained, width 90, nine attention heads, two blocks, where the graph regularizer is evaluated. Round 124 (w141) is the deletion-only graph; round 126 (w143) the whole-vector store affine on it, paired against round 124's control on that graph and against round 118's affine on the old one. Round 129 (w146) runs the deletion-only control for 300 epochs. Rounds 136 to 143 screen the essentiality finalists and the physically constrained stores of Section 3, every one at width 12 on the deletion-only graph over six seeds, paired against round 124's control and round 126's store: round 136 (w150) the vector with the six rule-count columns beside it, the essentiality study's best single model; round 137 (w151) the cached solve head, a hidden-256 perceptron over the store, in place of the affine; round 138 (w155) the linear minimization-of-metabolic-adjustment store; round 141 (w158) the max-min driving-force concentration store; round 143 (w160) the mean of six samples of each deletion's flux cone. Round 146 (w163) reads the size question, width 6 with two heads against width 12, on the store arm. The stores are written on the prototroph over the source store's genes and reaction order, with the growth-ratio column unchanged, so each pair isolates the vector.

Round 118 is read. On the amino acids the whole-vector affine into the 19-output mean reads 0.025 (sd 0.015) against the control's 0.034 (sd 0.025): −0.009 (sd 0.018), above on two of six, and both sit under every 027 arm on the same split (0.068, 0.061, 0.048 at width 90). Two seeds of each arm stayed near a constant predictor (validation Pearson 0.01 to 0.02 at selection; one control selected at epoch 2 and scores 0), so the width-12 backbone that the essentiality endpoint tuned is under-sized or unstable for 19 targets at once, and the width arm of round 122 is the test that has to be read before any amino-acid affine is. Averaged over seeds the control reads 0.058 and the store arm 0.035, so the affine costs a little on this target rather than adding to it.

Round 124 is read (job 2761, 3 h 38 min): the amino-acid control on the deletion-only graph, the Mülleder strain without the two alleles and the BTS1 copy it never carried. It reads 0.049 (sd 0.007) against the old-graph control's 0.034 (sd 0.025), +0.015 paired (sd 0.026), above on five of six. The level is modest; the spread is the finding. Every seed of the corrected graph escapes the constant predictor and reads 0.042 to 0.059, where two seeds of the old graph sat at 0.000 and 0.010, so the collapse round 118 attributed to the width-12 backbone was in part the graph carrying three genes the strain does not have. Ensembled over six seeds it reads 0.062 against 0.058, under 027's best at 0.068 and above its context-only cell at 0.061, both at width 90. The amino-acid affine arms of rounds 118 and 121 are paired against the old graph; the corrected control is the one any further affine on this target is paired against.

Round 121 is read (jobs 2755 and 2756, 4 h 34 min and 4 h 35 min): the amino-acid production store, on the old graph. The 19 synthesis ratios with the growth ratio read 0.031 (sd 0.004) against the control's 0.034, −0.003 paired (sd 0.025), above on two of six; the growth vector with all 42 summary columns reads 0.023 (sd 0.009), −0.011 paired (sd 0.027), above on two of six. Four of the 20-column arm's seeds were selected at validation Pearson 0.005 to 0.025 and still read 0.025 to 0.037 on test, the selection blindness of round 117 on a target where the affine has 19 outputs to fill from 20 columns. Ensembled over six seeds the two arms read 0.030 and 0.026 against the control's 0.058, so neither production summary carries rank the control does not, and the amino-acid affine has now been tried three ways on the old graph (the growth vector, 20 summaries, the vector with 42 summaries) without a gain. Round 126 is the same whole-vector affine on the corrected graph, against round 124's control.

Round 126 is read (job 2770, 3 h 40 min): the amino-acid whole-vector affine on the corrected graph, the first amino-acid affine paired on the strain as built. It reads 0.040 (sd 0.005) against the corrected-graph control's 0.049 (sd 0.007), −0.009 paired (sd 0.011), above on one of six, and ensembled 0.043 against 0.062. Against the old-graph affine of round 118 it reads +0.015 (sd 0.014), above on six of six, which is the size of the graph correction round 124 measured on the control, so the graph and not the store is what moved. The affine does on this target what it did on betaxanthin (the sibling document) to the seed spread, sd 0.005 against 0.007, and nothing to the level. Four forms of the amino-acid store have now been read, the growth vector on either graph and the two production summaries, against two controls, and none is above its control; the amino-acid module has no gain to report at width 12, and round 122's wide control is what says whether width changes that.

The amino-acid half of round 122 is read (jobs 2774 and 2775, 7 h 36 min and 7 h 48 min): the affine-free control on 027's backbone, width 90, on the old graph, the cell round 118 said had to be read before any amino-acid affine. It reads 0.034 (sd 0.010) against the width-12 control's 0.034 (sd 0.025), +0.000 paired (sd 0.031), above on three of six, and ensembled over six seeds 0.050 against 0.058, under the corrected-graph control's 0.062 at width 12 and under every 027 arm at this width (0.068, 0.061, 0.048). Every wide seed leaves the constant predictor, selected at epochs 10 to 59 with validation Pearson 0.09 to 0.13, and the validation loss bottoms at epochs 6 to 21 and rises 0.05 to 0.07 by epoch 199. So width does for the amino acids what the graph correction did (round 124): it removes the collapsed seeds and tightens the spread (sd 0.010 against 0.025), and it does nothing to the level. The 027 cells of this form read 0.061 to 0.068 over three seeds on the same split; the six seeds here read 0.016 to 0.043, and the difference between the two runs of one architecture is as large as any arm effect on this target. With the second amino-acid screen agreeing with the first at 0.03 and the first's own retest at 0.20 (Section 3), the amino-acid endpoint's strain-level rank has no headroom to show an arm effect against, and the width read closes the amino-acid backbone question rather than opening one.

Round 129 is read (job 2781, 5 h 34 min): the deletion-only amino-acid control run for 300 epochs instead of 200, the check that the 200-epoch budget is not cutting the amino-acid arms short. It reads 0.049 (sd 0.009) against the 200-epoch control's 0.049 (sd 0.007), −0.001 paired (sd 0.003), above on one of six, and ensembled over six seeds 0.060 against 0.062. Four of the six seeds select the same epoch with the same score in both runs (148, 88, 75 and 37), since the first 200 epochs of one seed are one trajectory; the two whose validation peak moved past epoch 200 (to 287 and 217) read 0.036 against 0.042 and 0.055 against 0.054. The validation loss bottoms at epochs 20 to 169 and drifts up by at most 0.014 over the remaining epochs. A hundred more epochs change nothing on this target, and the 200-epoch budget stands.

Round 136 is read (job 2786, 3 h 40 min): the essentiality study's best single store, the parsimonious vector with the six rule counts (0.934 there), in the affine slot on the amino-acid deletion-only graph. It reads 0.033 (sd 0.009) against the affine-free control's 0.049 (−0.016 paired, sd 0.012, above on one of six) and against the plain store affine's 0.040 (−0.007, sd 0.013, two of six), ensembled over six seeds 0.037 against 0.062 and 0.043. The seeds select late (epochs 47 to 186, five of six past 100) on validation Pearson 0.06 to 0.09, so the arm neither collapses nor escapes further than the control does. The rule counts that separate essential from non-essential genes carry nothing for the amino-acid rank, and the store affine's small loss on this target (round 125) is not recovered by them.

Round 137 is read (job 2787, 3 h 38 min): the cached solve head, a hidden layer of 256 on the store vector in place of the zero-initialized affine, on the amino-acid deletion-only graph. It reads 0.053 (sd 0.008) against the affine-free control's 0.049 (+0.004 paired, sd 0.011, above on four of six) and against the store affine's 0.040 (+0.013, sd 0.008, six of six), ensembled over six seeds 0.063 against 0.062 and 0.043. The seeds select at epochs 24 to 192 on validation Pearson 0.07 to 0.10 with the validation loss bottoming at 25 to 105. So the head recovers on every seed what the affine lost on this target and lands at the control, the highest single amino-acid arm of this study (0.033 to 0.049 for the others) and inside the 027 band (0.049 to 0.062). The store is not harmful on the amino acids when it is read nonlinearly; it is also not helpful, and the ensemble says the same.

Round 138 is read (job 2793, 3 h 40 min): the linear minimization-of-metabolic-adjustment store, the nearest feasible flux to the parent's parsimonious flux under each knockout, in the affine slot on the deletion-only graph. It reads 0.052 (sd 0.006) against the affine-free control's 0.049 (+0.003 paired, sd 0.011, above on four of six) and against the parsimonious store's 0.040 (+0.012, sd 0.008, five of six), ensembled over six seeds 0.056 against 0.062 and 0.043. The seeds select at epochs 86 to 197 on validation Pearson 0.07 to 0.09, none collapsed. So a store anchored to one wild-type vector, which Section 3 measures as almost free of the vertex degeneracy (11 no-effect rows differ from the wild type's

against the parsimonious store's 324), does not carry the parsimonious store's loss on this target: the third form of the flux vector read here, after the cached solve head (round 137), that undoes it, and none rises above the control.

Round 141 is read (job 2796, 3 h 41 min): the metabolite side in the affine slot, the max-min driving-force concentrations of Section 3 on the prototroph (2,806 metabolites in mM with three driving-force statistics, under the 100 kJ/mol cap), on the deletion-only graph. It reads 0.045 (sd 0.005) against the affine-free control's 0.049 (−0.005 paired, sd 0.007, above on two of six) and against the parsimonious store's 0.040 (+0.004, sd 0.009, five of six), ensembled over six seeds 0.050 against 0.062 and 0.043. The seeds select at epochs 32 to 146 on validation Pearson 0.05 to 0.07, none collapsed. So the concentration vector sits between the flux vector and the control, inside the seed spread of both, as every store does on this target; the second law adds a representation at the floor's level, as it added one at the flux store's level on betaxanthin (the sibling document, round 140).

Round 146 is read (job 2800, 3 h 14 min): the size question from the small side, the store arm on the deletion-only graph at width 6 with two attention heads, paired against the width-12 store. It reads 0.025 (sd 0.018) against 0.040 (−0.015 paired, sd 0.017, above on one of six), ensembled over six seeds 0.046 against 0.043. Three seeds sit near the constant predictor, selected at epochs 88 to 140 on validation Pearson 0.01 to 0.03 and reading 0.003 to 0.015; the three that escape select at 104 to 192 on 0.06 to 0.12 and read 0.035 to 0.049. So half the width brings back the collapse the corrected graph had removed (round 124) on half the seeds, and width 12 is the floor at which nineteen outputs train on this backbone: with the width-90 read of round 122 the size axis is read from 6 to 90, the level does not rise on it, and the small end loses seeds.

Round 148 is read (job 2802, 3 h 38 min): the probabilistic metabolite side in the affine slot, the mean concentration over six draws of each deletion's driving-force polytope (Section 3) with the point store's statistics and the count of constrained metabolites, on the deletion-only graph. It reads 0.032 (sd 0.008) against the affine-free control's 0.049 (−0.017 paired, sd 0.011, above on no seed), against the parsimonious store's 0.040 (−0.008, sd 0.012, one of six) and against the driving-force point's 0.045 (−0.012, sd 0.009, no seed), ensembled over six seeds 0.035 against 0.062, 0.043 and 0.050. The seeds select at epochs 84 to 184 on validation Pearson 0.04 to 0.09, none collapsed. So the region's centroid is the lowest of the three metabolite-side reads on this target and under every control on every seed, where on betaxanthin it was the highest ensemble (the sibling document, round 147): 2,810 columns whose draws carry the prototroph's region but not this rank cost the affine what the ladder read of them (0.016 by ridge), and the metabolite side on the amino acids ends where the flux side did, at or under the control.

Round 143 is read (job 2803, 3 h 40 min): the sampled cone in the affine slot, the mean of six OptGP draws of each deletion's flux cone at 99 percent of its optimum (Section 3), the Flux Cone Learning representation on this backbone, on the deletion-only graph. It reads 0.019 (sd 0.007) against the affine-free control's 0.049 (−0.030 paired, sd 0.007, above on no seed) and against the parsimonious store's 0.040 (−0.021, sd 0.010, no seed), ensembled over six seeds 0.023 against 0.062 and 0.043, the lowest arm of the study. The seeds select at epochs 104 to 196 on validation Pearson 0.05 to 0.07, none collapsed, and the validation loss bottoms at epochs 3 to 21, so the arm spends nearly all of training past its minimum. So the cone's mean costs the affine more than any other store on this target, on every seed against both controls, as it cost the most on betaxanthin (the sibling document, round 142) and read at the ladder's floor here (−0.004 by ridge): a deletion's feasible region averaged over six draws is the weakest representation of the flux family on both targets.

Round 153 is read (job 2808, 3 h 34 min): the GO multi-hot in the affine slot, 2,362 propagated terms over every annotated gene, into the 19-output head on the deletion-only graph, the annotation the backbone never saw. It reads 0.058 (sd 0.001 over seeds) against the affine-free control's 0.049 (+0.009 paired, sd 0.007, above on five of six) and against the parsimonious store's 0.040 (+0.018, sd 0.005, six of six), ensembled over six seeds 0.059 against 0.062 and 0.043: the highest single-seed transformer arm on this target, above the cached solve head's 0.053, with the first paired gain over the control that holds on five seeds of six. Every seed selects at epoch 2 or 3, where the validation Pearson peaks at 0.11 before the backbone has moved, and the validation loss bottoms at epochs 1 to 2 and rises by 0.12 to 0.13 by epoch 199: as on betaxanthin (the sibling document, round 150), the affine's read of the binary columns is the whole cell and the seeds read the same number. So annotation in the affine slot lifts this target by a hundredth where every metabolic store sat at or under the control, to the level the GO perceptron read alone (0.054) and under annotation with sequence through a perceptron (0.065); against a strain-level retest of 0.20 the gain is real and small, and it is the first.

Round 157 is read (job 2811, 3 h 38 min): the size question from the regularization side, the store arm on the deletion-only graph with the transformer's dropout raised from 0.1 to 0.3, the lever that lifted betaxanthin's store arm by 0.024 (the sibling document, round 145). It reads 0.045 (sd 0.005) against the store at 0.1 at 0.040 (+0.005 paired, sd 0.009, above on five of six) and against the affine-free control's 0.049 (−0.005, sd 0.009, two of six), ensembled over six seeds 0.049 against 0.043 and 0.062. The seeds select at epochs 39 to 187 on validation Pearson 0.05 to 0.08 and read 0.038 to 0.051, a band a third as wide as betaxanthin's. So dropout gives the store arm back half of what the store costs against the control, on five seeds of six, and leaves it under the control: the lever moves this target in the same direction as betaxanthin at a fifth of the size, inside the spread, on a target whose whole transformer band spans 0.04 to 0.06.

Round 160 is read (job 2815, 3 h 37 min): the HiGHS parsimonious deletion vector with the GO multi-hot appended, 4,131 flux columns (the wild-type row for a gene outside the GEM) and 2,362 annotation columns in one affine over every annotated gene, into the 19-output head on the deletion-only graph, the one-model form that read 0.211 on betaxanthin (the sibling document, round 151). It reads 0.055 (sd 0.002) against GO alone at 0.058 (−0.003 paired, sd 0.002, above on no seed) and against the parsimonious store alone at 0.040 (+0.015, sd 0.004, six of six), ensembled over six seeds 0.057 against 0.059 and 0.043. Every seed selects at epoch 2 on validation Pearson 0.11 to 0.12, round 153's pattern, and reads 0.051 to 0.059. So where the flux vector added 0.009 to the annotation on betaxanthin, on the amino acids it takes 0.003 from it, on every seed, the same direction as every read the flux row has joined on this target: the annotation's small gain over the control stands, and the metabolic vector still has nothing to add to it here.

Round 162 is read (job 2817, 3 h 40 min): the same affine, the HiGHS vector with GO appended, with transformer dropout 0.3, on the deletion-only graph. It reads 0.055 (sd 0.002) against the same affine at dropout 0.1 at 0.055 (+0.000 paired, sd 0.001, above on four of six) and against the store alone at dropout 0.3 at 0.045 (+0.011, sd 0.007, six of six), ensembled over six seeds 0.056 against 0.057 and 0.049. Every seed selects at epoch 2 on validation Pearson 0.11 to 0.12 and reads 0.051 to 0.059, the seeds of round 160 to the third decimal. So the transformer's dropout does not enter the annotation affine's read on this target either (the sibling document, rounds 159 and 161), and the affine's margin over the store narrows from 0.015 at dropout 0.1 (round 160) to 0.011 at 0.3 only because the store arm moved (round 157): the affine over the annotation is what this target's best transformer arm is, and the backbone around it is not where its level is set.

Round 165 is read (job 2823, 3 h 40 min): the GO affine alone with transformer dropout 0.3, on the deletion-only graph. It reads 0.058 (sd 0.001) against the GO affine at dropout 0.1 at 0.058 (+0.000 paired, sd 0.001, above on four of six) and against the store alone at dropout 0.3 at 0.045 (+0.013, sd 0.005, six of six), ensembled over six seeds 0.059 against 0.059 and 0.049. Every seed selects at epoch 2 or 3 on validation Pearson 0.11 and reads 0.057 to 0.059. So the third dropout read on this target closes the question as the betaxanthin rounds 159 and 161 closed it there: the annotation affine, alone or with the flux vector, reads the same number at either dropout, and the target's best transformer arm is a linear read of the annotation that the backbone does not enter.

Round 168 is read (jobs 2826 and 2827, 3 h 37 min and 3 h 39 min): six more seeds, 2706 to 2711, of the GO affine and of its affine-free control, so this target's one paired gain is read on twelve. The arm reads 0.057 (sd 0.002) against the control's 0.048 (sd 0.010), +0.009 paired (sd 0.010), above on ten of twelve, where the first six gave +0.009 above on five of six: the gain holds at the same size on twice the seeds and its sign is now established on ten of twelve. The two arms differ in character more than in level. The annotation arm reads 0.053 to 0.059 over twelve seeds, a range of 0.007; the control reads 0.023 to 0.059, a range of 0.036, five times as wide, because its score depends on whether its seed leaves the plateau. That is what the ensembles show: averaging the twelve control seeds reads 0.067, above the twelve-seed annotation ensemble's 0.058 and above every single cell of either arm, while averaging the annotation's seeds gains nothing over one of them. So on this target the annotation lifts a single seed and seed-averaging lifts the control past it: the one paired gain the study has is real at 0.009, and it is smaller than what the control's own seed noise is worth when averaged away.

Round 174 is read (job 2833, 3 h 43 min): six more seeds of the flux-and-annotation arm, the HiGHS vector with GO in one affine (round 160), so it has the twelve-seed level and ensemble the annotation arm has had since round 168. Over twelve it reads 0.056 (sd 0.003, standard error 0.0007), the seeds 0.051 to 0.060, every one selecting at epoch 2 or 3; the annotation arm alone reads 0.057 (sd 0.002) on the same twelve, and the flux vector's contribution beside GO, paired by seed, is −0.002 (sd 0.003), above on two of twelve, where the first six gave −0.003 on none of six. The twelve-seed ensembles are 0.057 and 0.058. So the reading of round 160 holds at twice the seeds: on this target the flux vector takes a little from the annotation on almost every seed, where on betaxanthin it adds 0.012 on eleven of twelve.

Round 183 is read (job 2845, 3 h 36 min): the reverse direction in the transformer's affine, the strain's measured Cachera betaxanthin score appended to GO, 2,363 columns read standardized because the score is signed. It reads 0.053 (sd 0.010) against GO alone at 0.058 (−0.005 paired, sd 0.010, two of six) and against the control's 0.049 (+0.004, sd 0.013, five of six), ensembled 0.056; the seeds read 0.040 to 0.065 and select at epochs 2 to 166 on validation Pearson 0.056 to 0.077, where the GO affine under log1p selects at epochs 2 to 3 on 0.106 to 0.112. So the column the ladder's forest reads at 0.071 alone and 0.078 beside GO adds nothing to the annotation in the affine, as the flux vector added nothing (round 174). The pair is not clean: the GO affine it is read against encodes the multi-hot as log1p, and this arm's store, the score and the multi-hot together, is standardized as one table, so the annotation's own columns enter at a different scale and the halved validation correlation may be the encoding's, not the score's. GO alone standardized (round 189) is the control that separates the two, and the score alone in the affine, one column and no annotation, is round 184.

Round 184 is read (job 2846, 3 h 33 min): the strain's betaxanthin score alone in the affine, one standardized column and no annotation. It reads 0.060 (sd 0.015) against the score with GO at 0.053 (+0.006 paired, sd 0.020, four of six), against GO alone at 0.058 (+0.002, sd 0.016, three of six) and against the control's 0.049 (+0.010, sd 0.018, four of six); the six-seed ensemble reads 0.073, above the control's twelve-seed ensemble of 0.067 and the highest ensemble of any arm on this target. The seeds read 0.036 to 0.080 and select at epochs 30 to 151 on validation Pearson 0.098 to 0.123, the highest validation correlation of any arm, from one column, where the GO affine's 2,362 columns give 0.106

to 0.112 and the same column beside GO under the standardized encoding gave 0.056 to 0.077 (round 183). So the measured score is the one input the transformer reads at the annotation's level on its own and above it ensembled, and the annotation appended to it under one standardized table (183) reads under either alone: the encoding control (189) decides whether that is the score and the annotation failing to add or the multi-hot's scale under standardization. The spread, sd 0.015 against the GO affine's 0.001, is the widest of any affine arm on this target.

Round 189 is read (job 2855, 3 h 33 min): the encoding control, GO alone standardized instead of log1p. It reads 0.053 (sd 0.012) against the log1p GO affine's 0.058 (−0.005 paired, sd 0.013, two of six), ensembled 0.061; the seeds read 0.039 to 0.073 and select at epochs 2 to 196 on validation Pearson 0.036 to 0.075, where the log1p affine selects at 2 to 3 on 0.106 to 0.112. Against it the score with GO under the same encoding (round 183) is +0.001 (sd 0.013, two of six). So round 183's deficit was the encoding's: standardizing the multi-hot costs the annotation 0.005 and its early, tight selection, and the score appended to it under that encoding adds nothing. The score adds to the annotation in the forest (0.078 against 0.063) and not in the affine beside it, while alone in the affine it reaches the annotation's level from one column (184); the twelve-seed level of that arm is round 195.

Round 195 is read (job 2871, 3 h 30 min): six more seeds of the score alone in the affine. The second six read 0.053 (sd 0.006) against the first six's 0.060, and over twelve the arm reads 0.056 (sd 0.012), seeds 0.036 to 0.080; paired over the same twelve seeds it is −0.001 against GO alone (sd 0.011, four of twelve) and +0.008 against the control (sd 0.017, seven of twelve). The twelve-seed ensemble reads 0.072, above the control's twelve-seed ensemble of 0.067 and the annotation's 0.058: at twelve seeds the score alone is the annotation's single-seed level with six times its spread, and its ensemble stays the highest of any arm on this target.

Round 205 is read (job 2891, 3 h 31 min): the betaxanthin score alone through a GELU hidden layer of 256, the nonlinear read of one column. It reads 0.078 (sd 0.007) against the same column in the linear affine at 0.060 (+0.018 paired, sd 0.015, five of six) and against GO alone at 0.058 (+0.020, sd 0.007, six of six), ensembled 0.082; the seeds read 0.072 to 0.090 and select at epochs 47 to 197 on validation Pearson 0.11 to 0.13. So the hidden layer does on this target what it did on betaxanthin: the single measured column read nonlinearly is the highest single-seed read of any transformer arm on the amino acids, at the forest's level over GO with the score (0.078), above the annotation on every seed, with the highest ensemble of any arm. The score store covers 563 of the 564 pinned genes with a target, so no coverage indicator enters this read (0.080 on the covered genes).

Round 206 is read (job 2892, 3 h 34 min): Cooper's seventeen relative peaks through a GELU hidden layer of 256, the other amino-acid screen read as an input for Mülleder's nineteen. It reads 0.046 (sd 0.006) against GO alone at 0.058 (−0.012 paired, sd 0.006, zero of six), ensembled 0.048; the seeds read 0.037 to 0.052 and select at epochs 95 to 153 on validation Pearson 0.06 to 0.13. On the 495 pinned genes Cooper measured it reads 0.040 against the annotation's 0.053, and Cooper's coverage carries nothing on this target (the indicator reads 0.005), so the comparison is the values alone. The transformer reads the peaks at half the forest's 0.093 over the same columns and under the annotation on every seed, where the perceptron read them at 0.049 and 0.052: on this target the second screen's columns reach the trained heads at the perceptron's level and the forest is the only model that reads them well. The linear control is round 207 (w227).

Table 1: Every arm paired by seed against its control on the same split (Merzbacher's 639 genes pinned to test, the 027 protocol). The statistic is the mean per-amino-acid pinned-test Spearman over the 19 columns at the epoch selected on validation Pearson (centered five-epoch moving average). Columns: seeds paired, the arm and control means with their standard deviations over seeds, the mean paired difference with its standard deviation, the seeds on which the arm scored higher. Written by `target_notes_tex_tables.py` from `results/transfer_readout.json` (`target_transfer_readout.py`).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
whole-vector store affine against the control	6	0.0248 (0.0149)	0.0342 (0.0251)	−0.0094 (0.0180)	2/6	–
deletion-only cell graphs against the control	6	0.0492 (0.0070)	0.0342 (0.0251)	+0.0150 (0.0262)	5/6	–
19 synthesis ratios and the growth ratio against the control	6	0.0314 (0.0042)	0.0342 (0.0251)	−0.0028 (0.0248)	2/6	–
growth vector with all 42 summaries against the control	6	0.0230 (0.0085)	0.0342 (0.0251)	−0.0113 (0.0268)	2/6	–
width 90, two blocks, against the width-12 control	6	0.0343 (0.0099)	0.0342 (0.0251)	+0.0000 (0.0314)	3/6	–
deletion-only graphs with the store affine against deletion-only graphs	6	0.0402 (0.0051)	0.0492 (0.0070)	−0.0090 (0.0109)	1/6	–
deletion-only graphs with the store affine against the store affine	6	0.0402 (0.0051)	0.0248 (0.0149)	+0.0154 (0.0144)	6/6	–
vector with rule counts on deletion-only graphs against deletion-only graphs	6	0.0334 (0.0091)	0.0492 (0.0070)	−0.0158 (0.0116)	1/6	–
vector with rule counts on deletion-only graphs against the store affine there	6	0.0334 (0.0091)	0.0402 (0.0051)	−0.0068 (0.0130)	2/6	–
solve head (hidden 256) on deletion-only graphs against deletion-only graphs	6	0.0534 (0.0081)	0.0492 (0.0070)	+0.0042 (0.0108)	4/6	–
solve head (hidden 256) on deletion-only graphs against the store affine there	6	0.0534 (0.0081)	0.0402 (0.0051)	+0.0132 (0.0084)	6/6	–
linear MOMA store on deletion-only graphs against deletion-only graphs	6	0.0518 (0.0061)	0.0492 (0.0070)	+0.0026 (0.0105)	4/6	–
linear MOMA store on deletion-only graphs against the parsimonious store there	6	0.0518 (0.0061)	0.0402 (0.0051)	+0.0116 (0.0078)	5/6	–
driving-force concentration store on deletion-only graphs against deletion-only graphs	6	0.0445 (0.0050)	0.0492 (0.0070)	−0.0047 (0.0068)	2/6	–
driving-force concentration store on deletion-only graphs against the parsimonious store there	6	0.0445 (0.0050)	0.0402 (0.0051)	+0.0043 (0.0091)	5/6	–
deletion-only store at width 6 against width 12	6	0.0252 (0.0182)	0.0402 (0.0051)	−0.0150 (0.0171)	1/6	–
driving-force polytope mean on deletion-only graphs against deletion-only graphs	6	0.0324 (0.0082)	0.0492 (0.0070)	−0.0169 (0.0105)	0/6	–
driving-force polytope mean on deletion-only graphs against the parsimonious store there	6	0.0324 (0.0082)	0.0402 (0.0051)	−0.0079 (0.0118)	1/6	–

Table 2: Table 1, continued (part 2 of 3).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
driving-force polytope mean on deletion-only graphs against the driving-force point	6	0.0324 (0.0082)	0.0445 (0.0050)	−0.0121 (0.0085)	0/6	–
GO multi-hot in the affine slot on deletion-only graphs against deletion-only graphs	6	0.0578 (0.0010)	0.0492 (0.0070)	+0.0086 (0.0067)	5/6	–
GO multi-hot in the affine slot on deletion-only graphs against the parsimonious store there	6	0.0578 (0.0010)	0.0402 (0.0051)	+0.0176 (0.0047)	6/6	–
HiGHS store with GO appended against GO alone	6	0.0550 (0.0024)	0.0578 (0.0010)	−0.0028 (0.0017)	0/6	–
HiGHS store with GO appended against the parsimonious store alone	6	0.0550 (0.0024)	0.0402 (0.0051)	+0.0148 (0.0037)	6/6	–
HiGHS store with GO appended at dropout 0.3 against 0.1	6	0.0553 (0.0024)	0.0550 (0.0024)	+0.0004 (0.0007)	4/6	–
HiGHS store with GO appended at dropout 0.3 against the store at 0.3	6	0.0553 (0.0024)	0.0447 (0.0054)	+0.0106 (0.0071)	6/6	–
HiGHS store with GO appended against GO alone, twelve seeds	12	0.0557 (0.0026)	0.0572 (0.0019)	−0.0015 (0.0026)	2/12	–
measured betaxanthin score with GO against GO alone	6	0.0533 (0.0097)	0.0578 (0.0010)	−0.0045 (0.0097)	2/6	–
measured betaxanthin score with GO against the control	6	0.0533 (0.0097)	0.0492 (0.0070)	+0.0041 (0.0129)	5/6	–
the betaxanthin score through a hidden layer of 256 against the score in the linear affine	6	0.0778 (0.0065)	0.0595 (0.0151)	+0.0183 (0.0153)	5/6	–
the betaxanthin score through a hidden layer of 256 against GO alone	6	0.0778 (0.0065)	0.0578 (0.0010)	+0.0200 (0.0072)	6/6	–
measured betaxanthin score alone in the affine against GO alone, twelve seeds	12	0.0561 (0.0115)	0.0572 (0.0019)	−0.0011 (0.0113)	4/12	–
measured betaxanthin score alone in the affine against the control, twelve seeds	12	0.0561 (0.0115)	0.0482 (0.0097)	+0.0078 (0.0170)	7/12	–
GO alone standardized against GO alone under log1p	6	0.0527 (0.0124)	0.0578 (0.0010)	−0.0052 (0.0131)	2/6	–
measured betaxanthin score with GO against GO alone under the same standardized encoding	6	0.0533 (0.0097)	0.0527 (0.0124)	+0.0007 (0.0132)	2/6	–
measured betaxanthin score alone in the affine against the score with GO	6	0.0595 (0.0151)	0.0533 (0.0097)	+0.0062 (0.0204)	4/6	–
measured betaxanthin score alone in the affine against GO alone	6	0.0595 (0.0151)	0.0578 (0.0010)	+0.0017 (0.0155)	3/6	–

Table 3: Table 1, continued (part 3 of 3).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
measured betaxanthin score alone in the affine against the control	6	0.0595 (0.0151)	0.0492 (0.0070)	+0.0103 (0.0180)	4/6	–
GO multi-hot in the affine slot against the control, twelve seeds	12	0.0572 (0.0019)	0.0482 (0.0097)	+0.0089 (0.0097)	10/12	–
GO affine at dropout 0.3 against 0.1	6	0.0581 (0.0008)	0.0578 (0.0010)	+0.0003 (0.0007)	4/6	–
GO affine at dropout 0.3 against the store at 0.3	6	0.0581 (0.0008)	0.0447 (0.0054)	+0.0134 (0.0053)	6/6	–
deletion-only store at dropout 0.3 against 0.1	6	0.0447 (0.0054)	0.0402 (0.0051)	+0.0045 (0.0092)	5/6	–
deletion-only store at dropout 0.3 against deletion-only graphs	6	0.0447 (0.0054)	0.0492 (0.0070)	−0.0045 (0.0087)	2/6	–
deletion-only graphs for 300 epochs against 200	6	0.0485 (0.0087)	0.0492 (0.0070)	−0.0007 (0.0027)	1/6	–
sampled-cone mean store on deletion-only graphs against deletion-only graphs	6	0.0189 (0.0066)	0.0492 (0.0070)	−0.0303 (0.0070)	0/6	–
sampled-cone mean store on deletion-only graphs against the parsimonious store there	6	0.0189 (0.0066)	0.0402 (0.0051)	−0.0213 (0.0097)	0/6	–

Table 4: The pinned-test Spearman of the 027 rounds on the same split and statistic, read by `target_transfer_readout.py` from 027's result files: the mean over three seeds for a trained arm, the validation-selected cell for the linear ladder, and the Merzbacher comparison's RandomForest.

comparator	Spearman
mulleder19	
027 concentration readout without constraints (wave 7), three seeds	0.0677
027 context-only concentration control (wave 8), three seeds	0.0605
027 pooled arm (wave 7), three seeds	0.0479
bilinear ridge, validation-selected embedding (job 1911)	0.0450

3 Physically constrained stores on the prototroph, and the store read without the backbone ✗

The module so far carries mass balance and nothing else: one parsimonious vector per deletion. On the essentiality endpoint the store ladder (the essentiality document's round 113) built the same vector under enzyme ceilings, sampled turnover, GECKO pools and a minimization of metabolic adjustment, and none moved the score beyond the seed spread. None of those programs had been run on the prototroph on minimal medium, which covers every GEM gene rather than the labeled ones. On 2026-09-23 they were, on CPU beside the packs, with reads that need no card.

The second law as fixed directions admits no growth. The direction rule of the essentiality study's round 114 (a reaction whose standard energy on the union-external route is at least T below zero loses its reverse part, at least T above it its forward part; unpriced reactions keep the GEM box) leaves the prototroph infeasible at both thresholds, as it had on the essentiality model, with 84 fixed directions running against the parsimonious wild-type flux at 10 kJ/mol and 44 at 30 kJ/mol. No store is written for an infeasible parent, and the outcome is recorded (`results/target_physics_store_amino_acids_sign10.json`, `_sign30`). Standard energies at 1 M cannot be imposed as directions on this model; a thermodynamically consistent program needs the concentrations as variables, which is what the driving-force store below does with a linear program.

The stores that do exist. The linear MOMA store on the prototroph (1,161 genes) has 175 lethal deletions against the parsimonious store's 161, moves the growth ratio by more than 0.01 on 63 genes, and moves the $\log(1 + |v|)$ vector on 567 genes with a median relative distance of 0.0 to the parsimonious deletion vector. Among the 918 deletions that leave growth unchanged, the HiGHS parsimonious store's vector differs from its own wild type's for 324 (Table 8, from `store_degeneracy.py`) and the MOMA store's, anchored to one wild-type vector, for 11: the parsimonious optimum has alternative vertices for a third of the no-effect deletions on this model too, fewer than on the cassette strain (the sibling document). The max-min driving-force store (`target_driving_force_store.py`) solves, per deletion, for the metabolite concentrations between 1 μM and 10 mM that maximize the smallest driving force over the reactions the parsimonious flux carries, with transport and proton-translocating reactions unconstrained and, in the capped variant, reactions priced above 100 kJ/mol set aside; the sampled store (`target_sampled_flux_store.py`) draws six OptGP samples of each deletion's cone at 99 percent of its optimum and keeps the mean over draws and the per-reaction spread.

The store read without the backbone (Table 9; `target_store_baselines.py`, the 027 ladder's split, standardization and selection, with a random forest added, the Flux Cone Learning form on one point per deletion; nearest neighbors average every training row tied at the k-th similarity, predictions within 10^{-9} count as one rank, and the validation criterion is rounded to 10^{-6}, so two runs agree to every digit). The screen is genome-wide and the stores cover the GEM genes, so most training genotypes read the wild-type row and a gene's row is one vector with no graph, no sequence and no pathway. On the amino acids every flux or sequence representation under every model sits between −0.004 and 0.058 (the HiGHS store 0.022 by kNN, ProtT5 0.052 by forest, the two concatenated 0.058), where the 027 bilinear ridge read

0.045 and the transformer arms 0.049 to 0.062. The GO multi-hot on the same ladder reads 0.044 by ridge, 0.025 by nearest neighbors and 0.063 by forest, and with the HiGHS vector appended 0.046, 0.029 and 0.062, so the annotation that lifted betaxanthin's ladder to 0.204 lifts this one from 0.058 to 0.063, beside the GO perceptron's 0.054 and the GO affine's 0.058 (round 153): the annotation reaches the same band as every other input on this target. The sampled cone (1,161 deletions, 161 lethal, six OptGP draws each at 99 percent of optimum, 4.3 h on twelve processes after the model copy's simplex cycling was removed with presolve and the warm-up's warm starts kept) reads −0.004, 0.031 and 0.023 by ridge, kNN and forest, its mean a median 5.9 relative distance from the parsimonious point; round 143 (w160) is its transformer read. The driving-force polytope sampled per deletion (`target_concentration_polytope_store.py`, the method in the sibling document: the region below the max-min optimum by 1 kJ/mol, coordinate hit-and-run, six draws, the chain seeded from a hash of the polytope) has on the prototroph a wild-type region of 249 constrained reactions with 59 binding over 319 of the 2,778 variable metabolites, a median spread of $\ln c$ across draws of 2.08 over them against 2.62 for an unconstrained one, and 405 viable rows whose mean differs from the wild type's; its mean and spread stores read 0.016, 0.011 and 0.008 and 0.018, 0.009 and 0.001 by ridge, kNN and forest, the floor of the ladder. Round 148 (w165) is the transformer read of the polytope mean.

The annotation-only family, on the same split (Table 5; `target_go_models.py`, the method in the sibling document): DCell's hierarchy with a 19-output head on the root's state, and a perceptron with one hidden layer of 256 over the propagated GO multi-hot, over the HiGHS store row, over ProtT5 and over their concatenations, six seeds at 200 epochs, the epoch selected on validation per-feature Pearson. The GO perceptron reads 0.054 (sd 0.002 over seeds), ProtT5 0.054 (sd 0.008), and the two together 0.065 (sd 0.003), ensembled 0.069, the highest single-seed read of anything on this target (the transformer arms 0.033 to 0.053 at width 12, the 027 arms 0.049 to 0.062 at width 90); DCell's hierarchy reads 0.035 (sd 0.015). The store row alone reads 0.019, and appending it lowers every read it joins, GO from 0.054 to 0.042 and GO with ProtT5 from 0.065 to 0.045, the same sign the transformer's affine gave the parsimonious store (round 126). So on the amino acids the flux vector costs a perceptron what it costs the transformer, and annotation with sequence sets the level, 0.065 against a strain-level retest of 0.20 (Section 4).

Table 5: DCell's ontology hierarchy (the essentiality study's shared-input form, a linear head on the root's state, auxiliary regressions at alpha 0.3) and a perceptron over the propagated GO multi-hot, on the ladder's split (2,339 GO columns with training support), full batch, Adam at 10^{-3}, 200 epochs, the epoch selected by a five-epoch moving average of the validation per-feature Pearson. Columns: parameters, the single-seed read (the mean pinned-test Spearman over the 19 amino acids with the standard deviation over seeds), the six-seed ensemble scored the same way, and the selected epochs. Hierarchy rows are at subsystem width 8 unless marked width 20. Written by `target_notes_tex_tables.py` from `results/go_models_amino_acids.json` and its `_w20` companion (`target_go_models.py`).

model	parameters	single seed (sd)	ensemble	selected epochs
DCell-style hierarchy, shared gene input, root head	354,777	+0.035 (0.015)	+0.058	120, 114, 71, 156, 40, 34
perceptron over the GO multi-hot	610,835	+0.054 (0.002)	+0.055	17, 16, 18, 17, 17, 16
perceptron over the HiGHS parsimonious store row	1,062,675	+0.019 (0.006)	+0.027	49, 143, 76, 48, 191, 66
perceptron over ProtT5	267,283	+0.054 (0.008)	+0.060	16, 12, 14, 14, 15, 9
perceptron over GO with the store row	1,668,371	+0.042 (0.003)	+0.041	16, 19, 19, 19, 18, 16
perceptron over GO with ProtT5	872,979	+0.065 (0.003)	+0.069	13, 10, 12, 11, 8, 14
perceptron over GO, ProtT5 and the store row	1,930,515	+0.045 (0.006)	+0.050	8, 11, 12, 14, 10, 9

The strain's measured betaxanthin score through the same perceptrons, on CPU (Table 6; the GO-models script with the score as its store block, one hidden width per pass, six seeds, seconds a seed): the score alone reads 0.050, 0.054 and 0.055 at widths 16, 256 and 1,024, the two wider settings selecting at epoch 198 of 200; a 400-epoch rerun of those two settings reads 0.055 and 0.053, selecting at 398 and at 321 to 384, so the longer budget finds no higher peak and the score-alone perceptron's level is 0.055 at any width; GO with the score reads 0.056, 0.064 and 0.066, ensembled 0.066 at 1,024 with a band of 0.002 and selection at epochs 10 to 11. Against the GO perceptron alone (0.054) the score adds 0.012 at the widest setting, the same direction as the forest's 0.015 (Section 4) and the opposite of the transformer's affine, where beside GO under one encoding it adds 0.001 (round 189). So on this target too the score adds to the annotation in a head trained alone and not in the transformer's slot; the forest over the same columns, 0.078, stays the highest read of any model.

The other amino-acid screen read as an input is the highest of any model on this target. Cooper's seventeen relative peaks, the store the betaxanthin document's Section 3 also reads, predict Mülleder's nineteen concentrations on the ladder at 0.033, 0.057 and 0.093 by ridge, nearest neighbors and forest, against 0.063 for the forest over GO alone, 0.078 for GO with the betaxanthin score and 0.071 for that score alone. So the ordering of inputs on this target is measured-to-measured first, then a measured endpoint from another pathway, then the annotation, and the two amino-acid screens read each other better than either reads its own gene's annotation, on 842 test genotypes of which 102 carry the reference row because Cooper did not measure them. Through the perceptron (Table 7) the peaks alone read 0.049 at width 256 and 0.052 at 1,024, half the forest's 0.093 over the same columns and under the GO perceptron (0.054): the forest reads Cooper's columns as it reads the betaxanthin score, better than any trained head does. Beside GO the peaks add: in one hidden layer 0.064 and 0.065, and GO linear beside the peaks' hidden layer of 1,024 reads 0.068

(sd 0.003, ensemble 0.070), above GO with ProtT5 (0.065) and GO with the betaxanthin score (0.066), the highest single-seed read of any perceptron on this target. So the ordering the forest gives is the perceptron's too: the peaks alone under the annotation, and the peaks beside it above every other pair. The transformer's read of the same columns, through the linear affine and through a hidden layer of 256, is queued (rounds 206, 207).

Table 6: The strain's measured Cachera betaxanthin score (one column, standardized on the training rows) alone and with the propagated GO multi-hot, through one ReLU hidden layer of the stated width and a linear head over the nineteen amino acids, on the ladder's split, full batch, Adam at 10^{-3}, 200 epochs, the epoch selected by a five-epoch moving average of the validation per-feature Pearson, on CPU. Columns: hidden width, parameters, the single-seed mean pinned-test Spearman over the amino acids with the standard deviation over six seeds, the six-seed ensemble, the range of selected epochs. Written by `target_notes_tex_tables.py` from `results/go_models_amino_acids_bxscore_h<H>.json` (`target_go_models.py -target amino_acids -store SM_gem_bxscore -mlp-hidden H`).

model	width	parameters	single seed (sd)	ensemble	epochs
perceptron over the measured betaxanthin score	16	355	+0.050 (0.006)	+0.058	161 to 198
perceptron over GO with the measured betaxanthin score	16	38,211	+0.056 (0.009)	+0.066	51 to 182
perceptron over the measured betaxanthin score	256	5,395	+0.054 (0.003)	+0.054	198 to 198
perceptron over GO with the measured betaxanthin score	256	611,091	+0.063 (0.004)	+0.064	17 to 17
perceptron over the measured betaxanthin score	1024	21,523	+0.055 (0.003)	+0.055	198 to 198
perceptron over GO with the measured betaxanthin score	1024	2,444,307	+0.066 (0.002)	+0.066	10 to 11

Table 7: Measured stores through a perceptron predicting Mülleder's nineteen amino acids, on CPU: Cooper's seventeen relative peaks (store YPD_bx_cooperpool), each log1p then standardized on the training rows, alone, with the propagated GO multi-hot through the same hidden layer, and with GO through a linear map beside the store's hidden layer (two slots), one ReLU hidden layer of the stated width and a linear head, on the ladder's split, full batch, Adam at 10^{-3}, 200 epochs, the epoch selected by a five-epoch moving average of the validation per-feature Pearson. Columns: the store, the arm, hidden width, parameters, the single-seed mean pinned-test Spearman over the amino acids with the standard deviation over six seeds; the six-seed ensemble, the range of selected epochs. Written by `target_notes_tex_tables.py` from `results/go_models_amino_acids_*.json` (`target_go_models.py -store <store> -mlp-hidden H`).

store	model	width	parameters	single seed (sd)	ensemble	epochs
Cooper	perceptron over the store	256	9,491	+0.049 (0.006)	+0.053	115 to 198
Cooper	perceptron over GO with the store	256	615,187	+0.064 (0.002)	+0.065	17 to 21
Cooper	GO linear beside the store's perceptron (two slots)	256	54,464	+0.062 (0.004)	+0.064	28 to 45
Cooper	perceptron over the store	1024	37,907	+0.052 (0.006)	+0.056	62 to 198
Cooper	perceptron over GO with the store	1024	2,460,691	+0.065 (0.002)	+0.066	10 to 11
Cooper	GO linear beside the store's perceptron (two slots)	1024	82,880	+0.068 (0.003)	+0.070	30 to 50

Table 8: Among the deletions whose growth ratio is 1, how many carry a vector that differs from the store's own wild-type row (relative ℓ_2 above 10^{-6} in $\log(1+|v|)$), with the median and ninetieth percentile of that distance. Written by `target_notes_tex_tables.py` from `031-betaxanthin-module/results/store_degeneracy.json` (`store_degeneracy.py`, which reads every target store).

store	genes	no effect	differ	median	p90
parsimonious, HiGHS two-stage	1161	918	324	0.000	0.376
linear MOMA	1161	907	11	0.000	0.000

Table 9: Each deleted gene represented by one row of a store ($\log(1 + |v|)$ of the flux columns, standardized on the training genes; a gene the store does not cover reads the wild-type row), fit by the 027 ladder's ridge (rank-swept bilinear for the amino acids) and cosine nearest neighbors, every training row tied at the k-th similarity averaged in, and by a 300-tree random forest, the Flux Cone Learning form on one point. Selected on validation per-feature Pearson rounded to 10^{-6}; the cell is the mean pinned-test Spearman over the 19 amino acids. Written by `target_notes_tex_tables.py` from `results/store_baselines_amino_acids.json` (`target_store_baselines.py`).

representation	columns	ridge	kNN	forest
HiGHS parsimonious store	4131	+0.006	+0.022	+0.020
linear MOMA store	4131	+0.018	+0.034	+0.032
six gene-rule counts	6	+0.025	+0.032	+0.029
flux statistics of the parsimonious build	8	+0.015	+0.037	+0.046
driving-force concentrations, every priced reaction	2809	+0.013	+0.020	+0.019
driving-force concentrations, cap 100 kJ/mol	2809	+0.016	+0.014	+0.017
sampled cone, mean over draws	4131	-0.004	+0.031	+0.023
driving-force polytope, mean concentration over draws	2810	+0.016	+0.011	+0.008
driving-force polytope, mean with the spread of ln c	5616	+0.018	+0.009	+0.001
GO multi-hot (2,362 propagated terms)	2362	+0.044	+0.025	+0.063
HiGHS parsimonious store with GO appended	6493	+0.046	+0.029	+0.062
measured Cachera betaxanthin score of the strain (1 column, standardized)	1	+0.034	+0.026	+0.071
GO multi-hot with the measured betaxanthin score appended	2363	+0.044	+0.024	+0.078
measured Cooper 2010 amino-acid peaks of the strain (17 relative columns)	17	+0.033	+0.057	+0.093
ProtT5 (027's ladder)	1024	+0.035	+0.025	+0.052
HiGHS store with ProtT5	5155	+0.033	+0.050	+0.058

4 The second amino-acid screen ✗

The second amino-acid screen does not corroborate the first (Table 10; `amino_acid_dataset_concordance.py`). Cooper 2010 profiled the auxotrophic MATa collection after overnight growth in synthetic complete medium by capillary electrophoresis, 17 peaks each a linear ratio to its plate mean, four of them composites and lysine split in two; Mülleder 2016 measured 19 amino acids in mM on the prototrophic collection on minimal medium. Over the 4,051 strains both screens hold, the matched pairs correlate at -0.006 (alanine) to $+0.059$ (glycine, p 2×10^{-4}), with glutamate at 0.043 and aspartate at 0.031 the only others off zero; the top 5 percent of strains overlap at chance on every pair (the largest enrichment 2.2, lysine peak b, 22 of 202 against 10 expected); and across all 17 by 19 cells the median $|\rho|$ is 0.018 and the maximum 0.127, on an unmatched pair, so the matched cells do not stand out from the background. The two retest references bound this: Mülleder's 194 strains measured twice give a mean Spearman of 0.20 (Pearson on $\log_{10}$ 0.43, the 027 read), and Cooper's 47 duplicated genes, paired from the loader's ledger against the served rows, give 0.06 (log Pearson 0.10). The Cooper pairs are the screen's own within-study replicate agreement: the 47 genes the collection carries as two deletion strains, grown from fresh colonies and measured in the same run, listed on adjacent rows of the released table (each row already the mean of up to two traces, so the instrument's repeat precision on one extract is not recoverable). Their rows spread as widely as the whole table (the $\log_{10}$ ratio's sd over the 17 peaks 0.10 to 0.53 against 0.11 to 0.51, medians 0.29 and 0.28), so the low agreement is not a restricted range: a typical gene's released value is not reproduced by the same study's second measurement of it, and 0.05 is the reliability ceiling of a Cooper strain-level value, against Mülleder's 0.20 from the same kind of re-measurement. A cross-screen correlation cannot exceed roughly the geometric mean of the two, about 0.11, and the observed 0.00 to 0.06 is at or below it. Cooper's released table carries almost no strain-level signal, so it can neither validate the Mülleder target strain by strain nor serve as a second one; and the medium and the auxotrophy differ between the screens, a biological difference that this replicate agreement cannot separate from noise. What the read fixes is the amino-acid endpoint's position: its own retest rank is 0.20, the second screen agrees with it at 0.03, and every model above, transformer or ladder, sits between 0.00 and 0.06. The retest is the agreement of two single-culture measurements of one strain, so it bounds a second screen, not a model: a model scored against one measurement is bounded by the square root of the reliability, the share of a measurement's variance that is between strains (Table 11). On the Pearson scale over $\log_{10}$ concentrations that bound averages 0.62 over the nineteen amino acids, from 0.00 for asparagine and 0.13 for methionine, whose retest spread equals their spread across strains, to 0.85 for glutamine; from the Spearman retest of 0.20 the same rule gives about 0.45. The 237 injections of the quality-control sample put the instrument's spread at 0.03 to 0.13 in $\log_{10}$ against a single-culture spread of 0.16 to 0.27, so the retest noise is culture to culture, not analytical. The models sit at a tenth of that bound, not at it: the target is hard to learn from single-culture labels, and no input has found more than a tenth of what the measurement would let a model read.

Table 10: The two deletion-collection amino-acid screens on their 4,051 shared strains, matched on the collection's ORF name. Cooper's CE-LIF peaks (a linear ratio to the plate mean; composite peaks list their components) against Mülleder's LC-MS concentrations (mM): the Spearman over strains both measured, the Pearson of the two log-scaled quantities, the overlap of the top 5 percent of strains against the count expected by chance, and the two within-study replicate references, Mülleder's (194 strains measured twice, Spearman on log10) and Cooper's duplicated genes (two deletion strains of one gene in the released table). Written by `target_notes_tex_tables.py` from `results/amino_acid_dataset_concordance.json`.

Cooper peak	Mülleder	strains	Spearman	log Pearson	top 5%	Mülleder replicate	Cooper replicate
arginine	arginine	4044	−0.000	+0.032	11 / 10.1	0.20	0.07
threonine	threonine	3752	+0.007	+0.007	7 / 9.4	0.19	0.03
alanine	alanine	4048	−0.006	+0.015	11 / 10.1	0.15	-0.02
serine	serine	3756	+0.007	+0.022	15 / 9.4	0.29	0.00
glycine	glycine	4044	+0.059	+0.047	10 / 10.1	0.05	0.13
glutamate	glutamate	4021	+0.043	+0.050	10 / 10.0	0.30	0.01
aspartate	aspartate	3736	+0.031	+0.035	11 / 9.4	0.22	0.04
lysine a	lysine	4045	+0.021	+0.043	19 / 10.1	0.10	0.17
lysine b	lysine	4032	+0.029	+0.057	22 / 10.1	0.10	0.02
leucine+isoleucine+citrulline	leucine+isoleucine	4043	+0.019	+0.025	15 / 10.1	0.23	0.12
leucine+isoleucine+citrulline	leucine	4043	+0.007	+0.025	13 / 10.1	0.24	0.12
leucine+isoleucine+citrulline	isoleucine	4043	+0.028	+0.021	18 / 10.1	0.22	0.12
glutamine+valine	glutamine+valine	4048	+0.009	−0.010	12 / 10.1	0.28	0.10
glutamine+valine	glutamine	4048	+0.009	−0.015	12 / 10.1	0.34	0.10
glutamine+valine	valine	4048	+0.013	+0.035	11 / 10.1	0.21	0.10
methionine+proline	methionine+proline	4037	+0.010	+0.032	17 / 10.1	0.24	0.26
methionine+proline	methionine	4037	+0.016	+0.022	6 / 10.1	0.21	0.26
methionine+proline	proline	4037	+0.006	+0.029	18 / 10.1	0.27	0.26
asparagine+tyrosine	asparagine+tyrosine	4044	+0.031	+0.024	8 / 10.1	0.12	-0.12
asparagine+tyrosine	asparagine	4044	+0.011	+0.006	14 / 10.1	0.18	-0.12
asparagine+tyrosine	tyrosine	4044	+0.029	+0.023	8 / 10.1	0.07	-0.12

Table 11: Mülleder 2016's retest, the 194 strains measured twice, per amino acid. Pearson and Spearman between the two $\log_{10}$ measurements; the quality-control sample's sd over its 237 injections (the instrument); the single-measurement sd from the pairs and the sd across strains (both $\log_{10}$); the reliability, the between-strain share of a measurement's variance; and its square root, the Pearson a perfect model of the true value would reach against one measurement. The retest correlation bounds a second screen; the square root of the reliability bounds a model. Written by `target_notes_tex_tables.py` from the 027 read `results/mulleder_retest_reliability.json`.

amino acid	pairs	Pearson	Spearman	QC sd	single sd	across sd	reliability	ceiling
alanine	193	0.28	0.15	0.03	0.16	0.21	0.40	0.64
aspartate	193	0.46	0.22	0.03	0.16	0.22	0.43	0.66
glutamate	194	0.67	0.30	0.05	0.21	0.36	0.66	0.82
phenylalanine	194	0.46	0.16	0.05	0.21	0.29	0.45	0.67
glycine	190	0.12	0.05	0.04	0.19	0.21	0.20	0.45
histidine	194	0.50	0.26	0.13	0.18	0.26	0.49	0.70
isoleucine	194	0.52	0.22	0.05	0.21	0.30	0.52	0.72
lysine	194	0.48	0.10	0.07	0.18	0.23	0.43	0.65
leucine	194	0.49	0.24	0.05	0.22	0.30	0.47	0.69
methionine	189	0.19	0.21	0.05	0.27	0.27	0.02	0.13
asparagine	190	0.37	0.18	0.04	0.22	0.22	0.00	0.00
proline	194	0.64	0.27	0.04	0.22	0.33	0.57	0.76
glutamine	194	0.67	0.34	0.05	0.23	0.43	0.72	0.85
arginine	194	0.37	0.20	0.12	0.26	0.36	0.50	0.71
serine	194	0.54	0.29	0.05	0.21	0.31	0.55	0.74
threonine	194	0.56	0.19	0.03	0.21	0.31	0.55	0.74
valine	194	0.48	0.21	0.04	0.22	0.31	0.51	0.72
tryptophan	190	0.24	0.14	0.06	0.19	0.22	0.24	0.49
tyrosine	193	0.20	0.07	0.13	0.25	0.30	0.32	0.57
mean		0.43	0.20				0.42	0.62

Cooper's peaks as a target confirm the reliability read from the model side. The 17 peaks, read from the loader's table by the record's deleted gene on the same pinned split every other target uses (816 test genotypes carry a Cooper row, 561 of them pinned), fit by the ladder's ridge, nearest neighbors and forest over the same representations as the Mülleder target (Table 12), read between −0.025 and +0.041 mean Spearman over the peaks, the two highest the forest over the HiGHS store with ProtT5 (0.041) and the ridge over the adjustment store (0.039), against a Mülleder ladder of 0.00 to 0.06. The flux stores are the prototroph's on minimal medium, not Cooper's auxotroph on complete medium, so the metabolic rows are a mismatched input here; the annotation and the sequence, which do not depend on the medium, read 0.019 and 0.028 at best. A target whose own duplicate strains agree at 0.06 gives no model more than 0.04 to find.

Table 12: Each deleted gene represented by one row of a store ($\log(1 + |v|)$ of the flux columns, standardized on the training genes; a gene the store does not cover reads the wild-type row), fit by the 027 ladder's ridge (rank-swept bilinear for the amino acids) and cosine nearest neighbors, every training row tied at the k-th similarity averaged in, and by a 300-tree random forest, the Flux Cone Learning form on one point. Selected on validation per-feature Pearson rounded to 10^{-6}; the target is Cooper 2010's 17 CE-LIF peaks (a linear ratio to the plate mean, the auxotroph on synthetic complete medium), read from the loader's table by the record's deleted gene on the same pinned split (561 pinned test genotypes carry a Cooper row), over the SM stores of the prototroph; the cell is the mean pinned-test Spearman over the 17 peaks. Written by `target_notes_tex_tables.py` from `results/store_baselines_cooper.json` (`target_store_baselines.py`).

representation	columns	ridge	kNN	forest
HiGHS parsimonious store	4131	-0.010	-0.007	-0.003
linear MOMA store	4131	+0.039	+0.004	+0.010
six gene-rule counts	6	-0.006	+0.013	+0.003
flux statistics of the parsimonious build	8	-0.011	-0.016	-0.010
driving-force concentrations, every priced reaction	2809	+0.001	+0.018	-0.001
driving-force concentrations, cap 100 kJ/mol	2809	+0.014	+0.020	+0.019
sampled cone, mean over draws	4131	-0.014	-0.020	+0.008
driving-force polytope, mean concentration over draws	2810	-0.022	-0.012	+0.006
driving-force polytope, mean with the spread of ln c	5616	-0.025	-0.016	+0.006
GO multi-hot (2,362 propagated terms)	2362	-0.002	+0.003	+0.019
HiGHS parsimonious store with GO appended	6493	-0.002	+0.012	+0.019
ProtT5 (027's ladder)	1024	+0.029	+0.001	+0.005
HiGHS store with ProtT5	5155	+0.015	+0.020	+0.041

The reverse direction. The betaxanthin study read the strain's measured amino-acid pool as an input and found the ladder's forest carrying it at 0.280 (its Section 3); the same store form carries the strain's Cachera betaxanthin score, one signed column standardized on the training strains, into this target on the same pinned split (Table 9). Alone it reads 0.034 by ridge, 0.026 by nearest neighbors and 0.071 by forest, mean Spearman over the nineteen amino acids; appended to GO, 0.044, 0.024 and 0.078. The forest over one measured column is above every genotype-side representation on this target, and with GO it is the highest read of any model, transformer or ladder, 0.078 against the forest over GO alone at 0.063 and the GO affine's 0.058. The per-amino-acid reads of the column alone follow the cross-endpoint correlations (aspartate 0.185, threonine 0.125, tyrosine 0.084 by ridge). As on betaxanthin, this is phenotype to phenotype and it says the shared strain-level signal transfers in both directions; on a target whose annotation gain is 0.009, one measured column is worth more than the annotation.

4.1 The betaxanthin endpoint against the amino-acid pool ✗

The three screens profile the same deletion collection, so a gene both screens hold is one paired observation of the two endpoints, and whether a joint model of betaxanthin and the amino acids has anything to share is a question the data answer before any model is trained (Figure 1, Table 13; `betaxanthin_amino_acid_correlation.py`). Over the 4,477 strains Cachera and Mülleder both measured, the betaxanthin score correlates with methionine at +0.130 (bootstrap interval 0.100 to 0.157), with threonine at +0.093 and aspartate at +0.091, and against tyrosine at −0.083, arginine at −0.077 and glutamate at −0.065; the top 5 percent of strains on the two axes overlap at 3.3 times chance for aspartate (37 of 224 against 11.2) and 3.0 times for threonine, and at chance for the rest. Each of these sits well under its ceiling, the correlation two perfectly related quantities would show through the betaxanthin score's reliability of 0.84 and the amino acid's retest (0.20 to 0.54 over the nineteen), so the single-amino-acid reads are weak but not at their floor. The pool read is the one that matters for a joint model: a ridge from the nineteen standardized log concentrations to the betaxanthin score, fit and scored out of fold over five strain folds, reads a Spearman of +0.226 (folds 0.182 to 0.272) against −0.002 (sd 0.016) with the target shuffled. A deletion's measured amino-acid pool carries about as much rank on its betaxanthin level as the best genotype-side model in the betaxanthin study reads from the deleted gene (0.211, `notes-tex/031-betaxanthin-module`, round 151), on a different set of strains and from a different kind of input, so the two endpoints share strain-level signal well above the noise. Whether the pool adds to the deleted gene or restates it is read on the same strains and folds: the same ridge from the gene's propagated GO multi-hot (the betaxanthin study's best genotype-side input, 2,362 columns, an all-zero row for the 557 strains outside the store) reads 0.165, and from GO with the pool appended, the GO block scaled to the pool block's variance so one penalty serves both, 0.237 (folds 0.207 to 0.311), above the pool alone on four folds of five and above GO alone on every fold. The pool adds 0.072 beyond the gene's identity, so the amino-acid signal a joint model would learn from is not the one the annotation already carries. The Cooper screen gives the same picture faintly: no single peak beyond ±0.063 over the 4,225 shared strains, a pool ridge at +0.083 against a shuffled −0.003, with the peaks' own duplicate agreement bounding the ceilings at 0.00 to 0.49; on these strains GO alone reads 0.193 and GO with the Cooper pool 0.196, an addition of 0.003. Hypothesis (untested): part of the shared signal is a growth or respiratory covariate common to both screens, since the betaxanthin screen's top hits are mitochondrial and the pool responds to respiratory deficiency.

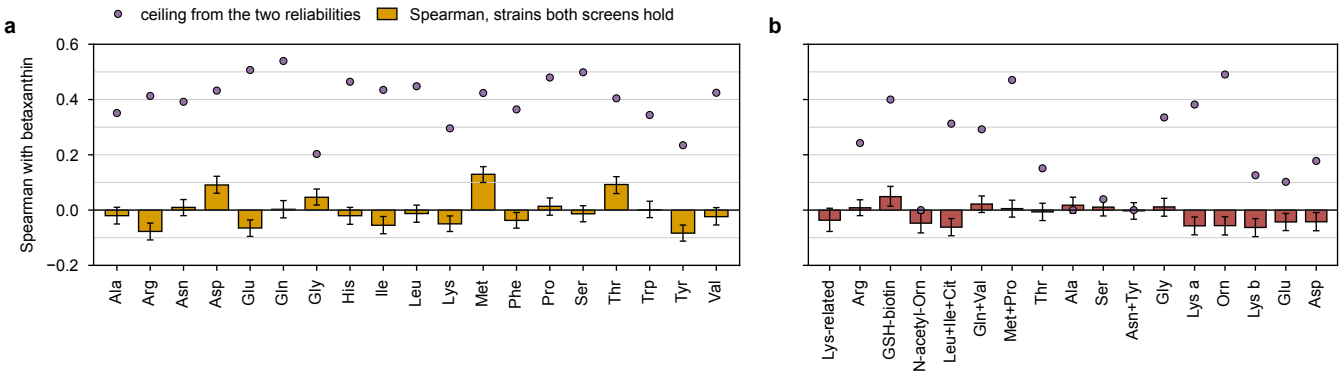


Figure 1: The betaxanthin score against each amino-acid quantity, strain by strain. a, Cachera 2023's betaxanthin score against each of Mülleder 2016's nineteen concentrations over the 4,477 strains both screens hold: the Spearman (bars) with a strain-bootstrap 95 percent interval (1,000 draws) and, above each bar, the ceiling a correlation could reach through both endpoints' noise, the geometric mean of the betaxanthin reliability (0.84) and the amino acid's log10 retest Spearman. **b**, The same against Cooper 2010's seventeen peaks over the 4,225 shared strains, the ceiling from Cooper's duplicate-strain agreement (absent where no duplicate pairs were measured). Written by `betaxanthin_amino_acid_correlation.py`.

Table 13: Cachera 2023's betaxanthin score (corrected 24 h colony fluorescence, population-centered) against each of Mülleder 2016's amino-acid concentrations, over the strains both screens hold, matched on the systematic ORF: the Spearman with a strain-bootstrap 95 percent interval (1,000 draws), the overlap of the top 5 percent of strains on the two axes against the count expected by chance, and the ceiling a correlation could reach through both endpoints' noise, the geometric mean of the betaxanthin reliability (0.84, the 027 variance-components read) and the amino acid's log10 retest Spearman over 194 strains. The ridge rows predict the betaxanthin score over five strain folds, the alpha chosen on an inner split, scored out of fold, with the same fit against a shuffled target: from the standardized log profile (the pool), from the deleted gene's propagated GO multi-hot (a zero row outside the store), and from both, the GO block scaled to the pool block's total variance; for these rows the interval column holds the lowest and highest fold and the top-5 column the shuffled-target fit's mean (sd), with the column count in parentheses. Written by `target_notes_tex_tables.py` from `results/betaxanthin_amino_acid_correlation.json` (`betaxanthin_amino_acid_correlation.py`).

Mülleder 2016 (mM)	strains	Spearman	95% interval	top 5%	ceiling
alanine	4,477	−0.020	[−0.050, +0.010]	24 / 11.2	0.35
arginine	4,477	−0.077	[−0.108, −0.047]	4 / 11.2	0.41
asparagine	4,477	+0.010	[−0.020, +0.038]	23 / 11.2	0.39
aspartate	4,477	+0.091	[+0.061, +0.122]	37 / 11.2	0.43
glutamate	4,477	−0.065	[−0.095, −0.036]	3 / 11.2	0.51
glutamine	4,477	+0.003	[−0.028, +0.034]	5 / 11.2	0.54
glycine	4,477	+0.046	[+0.018, +0.076]	11 / 11.2	0.20
histidine	4,477	−0.020	[−0.051, +0.010]	7 / 11.2	0.46
isoleucine	4,477	−0.055	[−0.086, −0.023]	17 / 11.2	0.43
leucine	4,477	−0.013	[−0.044, +0.018]	6 / 11.2	0.45
lysine	4,477	−0.050	[−0.078, −0.021]	9 / 11.2	0.30
methionine	4,477	+0.130	[+0.100, +0.157]	14 / 11.2	0.42
phenylalanine	4,477	−0.037	[−0.065, −0.009]	7 / 11.2	0.36
proline	4,477	+0.014	[−0.019, +0.044]	7 / 11.2	0.48
serine	4,477	−0.014	[−0.042, +0.016]	4 / 11.2	0.50
threonine	4,477	+0.093	[+0.060, +0.121]	34 / 11.2	0.40
tryptophan	4,477	+0.001	[−0.027, +0.032]	13 / 11.2	0.34
tyrosine	4,477	−0.083	[−0.112, −0.054]	23 / 11.2	0.23
valine	4,477	−0.024	[−0.054, +0.009]	17 / 11.2	0.42
ridge, pool (19)	4,477	+0.226	[+0.182, +0.272]	−0.002 (0.016)	
ridge, GO (2,362)	4,477	+0.165	[+0.136, +0.212]	+0.002 (0.012)	
ridge, GO with pool (2,381)	4,477	+0.237	[+0.207, +0.311]	+0.001 (0.014)	

Table 14: Table 13 for Cooper 2010's peaks (a linear ratio to the plate mean), the ceiling from the peak's duplicate-strain Spearman, blank where no duplicate pairs were measured.

Cooper 2010 (ratio)	strains	Spearman	95% interval	top 5%	ceiling
lysine related peak1	1,962	−0.037	[−0.077, +0.007]	1 / 4.9	
arginine	4,217	+0.008	[−0.020, +0.037]	13 / 10.6	0.24
gshbiotin	2,772	+0.049	[+0.014, +0.086]	5 / 7.0	0.40
n acetylornithine	3,229	−0.047	[−0.083, −0.011]	9 / 8.0	0.00
leucine+isoleucine+citrulline	4,217	−0.062	[−0.093, −0.031]	6 / 10.6	0.31
glutamine+valine	4,222	+0.022	[−0.009, +0.051]	13 / 10.5	0.29
methionine+proline	4,211	+0.005	[−0.025, +0.036]	6 / 10.6	0.47
threonine	3,899	−0.007	[−0.038, +0.025]	13 / 9.8	0.15
alanine	4,221	+0.018	[−0.012, +0.047]	8 / 10.5	0.00
serine	3,910	+0.011	[−0.021, +0.042]	11 / 9.8	0.04
asparagine+tyrosine	4,216	−0.003	[−0.033, +0.027]	11 / 10.6	0.00
glycine	4,217	+0.012	[−0.022, +0.043]	13 / 10.6	0.34
lysine a	4,219	−0.057	[−0.090, −0.025]	12 / 10.6	0.38
ornithine	3,948	−0.056	[−0.090, −0.024]	7 / 9.8	0.49
lysine b	4,207	−0.063	[−0.096, −0.031]	14 / 10.5	0.13
glutamate	4,189	−0.043	[−0.074, −0.012]	14 / 10.4	0.10
aspartate	3,880	−0.042	[−0.075, −0.009]	9 / 9.7	0.18
ridge, pool (17)	4,225	+0.083	[+0.054, +0.137]	−0.003 (0.010)	
ridge, GO (2,362)	4,225	+0.193	[+0.140, +0.260]	−0.003 (0.014)	
ridge, GO with pool (2,379)	4,225	+0.196	[+0.158, +0.253]	−0.006 (0.014)	

4.2 The third endpoint: isobutanol ✗

The same read on Montano Lopez 2024's isobutanol biosensor, over the 4,360 strains it shares with Mülleder (the screen holds 4,554), with the fold change read as log10 as its own reliability was: the pool predicts the endpoint where the annotation does not. Ridge from the nineteen concentrations reads the biosensor out of fold at 0.102, against 0.017 from the deleted gene's GO multi-hot and 0.098 from the two together, with the five folds at 0.056 to 0.132 and permuted labels at −0.003. So the measured pool carries six times what the gene's annotation carries on this endpoint, and the annotation adds nothing beside it, the same ordering as on betaxanthin (0.226 against 0.165, with the pool adding 0.072 there).

One amino acid at a time the correlations are small and the branched-chain ones are not where they sit. The biosensor is a branched-chain reporter by construction, yet leucine reads −0.029, isoleucine −0.026 and valine +0.004, every interval covering zero; the largest is arginine at +0.093 (interval 0.063 to 0.124, top-5 enrichment 2.4), then glutamine at −0.075, aspartate −0.051, threonine −0.044 and proline −0.042. Seven of the nineteen intervals exclude zero. Against a mean ceiling of 0.24, set by the geometric mean of the screen's within-screen duplicate agreement (0.31 over 190 repeated strains) and each amino acid's retest, the pool's 0.102 is about forty percent of what the noise allows and no single concentration reaches a fifth of it.

Two readings follow, and only the first is measured. The signal the pool carries about the biosensor is spread over the profile rather than held by the amino acids the sensor reports, since no single concentration approaches the pool fit. Why the reported branched-chain pools are the uncorrelated ones is not established here; a biosensor responding to pathway flux rather than to the standing pool would produce it (hypothesis, untested), and so would the two screens' different media, Mülleder's prototroph on minimal medium against the screen's auxotroph on complete medium.

4.3 The endpoint matrix, and what a joint model could share ✗

The question behind the isobutanol read is whether a second production endpoint trained beside betaxanthin could lift it. What a joint model can share is bounded by the strain-level signal the two endpoints hold in common, and the third pair of the matrix closes it: Cachera's betaxanthin score against the biosensor over the 4,353 strains both screens hold reads −0.041 (interval −0.070 to −0.013) against a ceiling of 0.51, eight percent of what the two endpoints' noise allows and on the wrong side of zero. The full matrix, each pair read on the strains both screens measured, with the pool entries the out-of-fold ridge from the nineteen concentrations and every ceiling built by one rule (Table 15): the amino-acid pool shares 56 percent of its ceiling with betaxanthin and 42 percent with the biosensor, and the two production endpoints share 8 percent of theirs with each other.

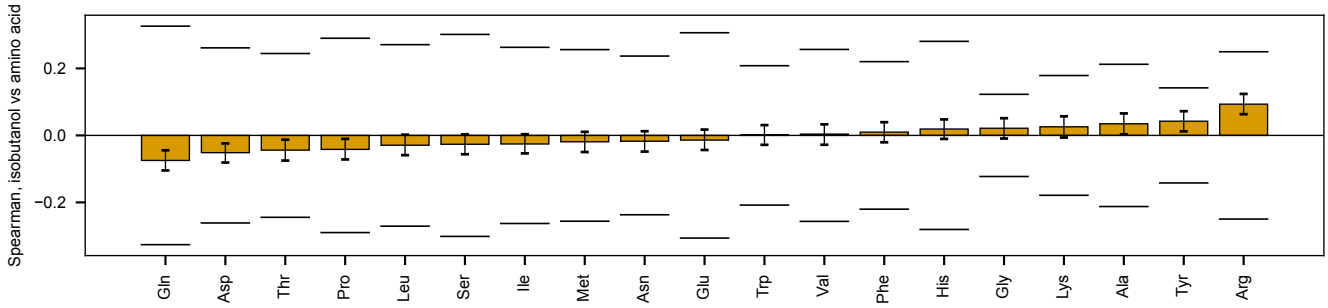


Figure 2: The isobutanol biosensor against each amino-acid concentration, strain by strain. Montano Lopez 2024's biosensor fold change (log10) against each of Mülleder 2016's nineteen concentrations over the 4,360 strains both screens hold, ordered by the correlation: the Spearman (bars) with a strain-bootstrap 95 percent interval (1,000 draws) and, as the rules above and below each bar, the ceiling a correlation could reach through both endpoints' noise, the geometric mean of the screen's within-screen duplicate agreement and the amino acid's log10 retest Spearman. Written by `isobutanol_amino_acid_correlation.py`.

Table 15: The strain-level signal each pair of endpoints holds in common, on the strains both screens measured, with the ceiling each read is bounded by. The two pool rows are the out-of-fold Spearman of a ridge from Mülleder's nineteen concentrations over five strain folds; the last row is the Spearman of Cachera's betaxanthin score against the biosensor's log10 fold change. Each ceiling is the geometric mean of the two endpoints' reliabilities, averaged over the amino acids where the pair involves the pool. Written by `target_notes_tex_tables.py` from `betaxanthin_amino_acid_correlation.json` and `isobutanol_amino_acid_correlation.json`.

pair	strains	shared signal	ceiling	share of ceiling
betaxanthin and the amino-acid pool	4,477	+0.226	0.40	56 percent
the amino-acid pool and isobutanol	4,360	+0.102	0.24	42 percent
betaxanthin and isobutanol	4,353	−0.041	0.51	8 percent

The one prior on the design is negative. In August a shared-trunk model of betaxanthin with the nineteen amino acids as an auxiliary head read −0.020 against betaxanthin alone, paired over four seeds with a spread of 0.012, and the auxiliary head tripled the run-to-run spread of the betaxanthin read, from 0.007 to 0.023, so detecting a lift of 0.015 on that design needs about twenty-two seeds an arm. Against a shared signal of −0.041 the isobutanol version of that design has less to share than the amino-acid one that failed, at the same cost. The other endpoint as an input rather than a second task is where the campaign's gains have come from, and that is the form the third endpoint should take if it is read at all: the amino-acid pool through a hidden layer as an input for the biosensor, which the ridge already reads at 0.102.

5 Where the module stands on this target ✗

The amino-acid endpoint is noisy, and every model reads at a tenth of what the noise allows. Its own strain-level retest is 0.20, the second screen agrees with it at 0.03, and every model, transformer or ladder, at either width, with or without a store, reads between 0.00 and 0.06. Four forms of the store affine sit at or under their control; the solve head recovers the affine's loss and lands at the control; width, 100 more epochs and the rule counts change nothing. What the campaign fixes is the target's position rather than the module's: the retest of 0.20 bounds a second screen, the bound on a model against one measurement is its square root, about 0.45 on the Spearman scale and 0.62 on the Pearson (Section 4), and every arm reads at a tenth of it, so the arms differ within a band that is small against the ceiling and against the seed spread alike. The physically stated stores read in the transformer confirm it from the other side: the adjustment store and the driving-force concentrations each sit between the parsimonious store and the control inside both spreads, so the parsimonious store's small loss is specific to that build and nothing metabolic rises above the control. The one read above the transformer band is annotation with sequence through a perceptron, 0.065, and the flux row lowers every read it joins. The sampled cone (round 143, 0.019) and the polytope mean (148, 0.032) read under every other store; width 6 (146, 0.025) brings the collapse back; dropout 0.3 on the store arm (157, 0.045) returns half of what the store costs and stays under the control. GO in the affine slot (153, 0.058) is the one transformer arm above its control on five seeds of six, at the GO perceptron's level; the HiGHS vector appended to it in the same affine takes 0.003 from it on every seed (160), where on betaxanthin it added 0.009, and dropout 0.3 around that affine moves it by less than a thousandth (162), as does dropout around the GO affine alone (165): as on betaxanthin, the annotation arm is the affine alone. On twelve seeds the annotation's gain holds at 0.009, above the control on ten of

twelve, but averaging the control's twelve seeds reads 0.067, above the annotation's twelve-seed ensemble and above every single cell of either arm (168): the gain is real and smaller than what the control's own seed noise is worth when averaged away. The flux vector beside GO on twelve seeds is −0.002, above the annotation alone on two of twelve (174): what adds 0.012 on betaxanthin takes on this target, at any seed count read. The one input that moves the target is measured, not inferred: the strain's own betaxanthin score, a single column, reads 0.071 by forest and 0.078 appended to GO, the highest read of any model on this target (Section 4). In the transformer's affine the same column beside GO reads 0.053, under the GO affine on four of six (183), with the pair confounded by the store's encoding; the column alone reads 0.060 with the highest validation correlation and, ensembled, the highest test rank of any arm on this target, 0.073 (184). The measured score is the one input the transformer reads at the annotation's level from a single column. The encoding control settles the pair: standardizing GO costs the affine 0.005 and its early selection, and beside GO under that encoding the score is +0.001 (189), so the score adds to the annotation in the forest and not in the affine. On twelve seeds the score alone holds the annotation's single-seed level (0.056 against 0.057) with six times its spread, and its twelve-seed ensemble, 0.072, stays the highest of any arm on this target (195). Through a hidden layer of 256 the same column reads 0.078 (sd 0.007), +0.018 over its linear read on five of six and +0.020 over the annotation on six of six, ensembled 0.082 (205): the highest single-seed transformer read and the highest ensemble on this target. The highest read of any model, though, is the other amino-acid screen: Cooper's seventeen peaks predict Mülleder's nineteen at 0.093 by forest, above GO with the score (0.078), the score alone (0.071) and GO alone (0.063), so the two screens read each other better than either reads its own gene's annotation (Section 3). The third endpoint agrees: the pool predicts the isobutanol biosensor out of fold at 0.102 against the annotation's 0.017 (Section 4.2). On all three targets a measured phenotype of the same strain is worth more than the gene's annotation. Through a perceptron Cooper's peaks alone read 0.052, under the annotation and half the forest, and GO linear beside them 0.068, the highest single-seed read of any perceptron on this target (Section 3). The transformer reads the same columns through a hidden layer of 256 at 0.046, under the annotation on every seed and half the forest's read of them (206), with the linear control queued (207): the forest is the only model that reads Cooper's peaks above the annotation.